

Student Manager App - Python Mini Project Report

1. Project Title

Student Manager App

2. Objective

To develop a simple Python application for managing student records, viewing student details, and checking pass/fail results based on marks.

3. Introduction

The Student Manager App is a menu-driven Python application that allows users to add student records, view stored records, and check examination results. It demonstrates the use of dictionaries, loops, conditional statements, and user input handling in Python.

4. Features

- Add Student
- View Student Records
- Check Student Result
- Exit Application

5. Technologies Used

Programming Language: Python

Data Structure Used: Dictionary

Concepts Used: Loops, Conditional Statements, User Input, Dictionary Operations

6. Algorithm

Step 1: Create an empty dictionary named student.

Step 2: Display menu options repeatedly using a loop.

Step 3: If user selects Add Student, accept name and marks and store them.

Step 4: If user selects View Student, display all student records.

Step 5: If user selects Check Result, verify marks and display Pass or Fail.

Step 6: If user selects Exit, terminate the program.

Step 7: Repeat until Exit is chosen.

7. Data Structure Explanation

A Python dictionary is used where the student name acts as the key and marks act as the value.

8. Working of the Program

Add Student: Stores name and marks.

View Student: Displays all saved student records.

Check Result: If marks are greater than or equal to 40, result is Pass; otherwise Fail.
Exit: Closes the application.

9. Sample Output

1. Add Student
2. View Student
3. Check Result
4. Exit

Enter student name: Rahul
Enter marks: 75
Rahul Added Successfully

Student Name: Rahul, Marks: 75, Result: Pass

10. Advantages

- Easy to understand and use.
- Demonstrates basic Python concepts.
- Can be extended to store more student information.
- Useful for beginners learning Python.

11. Limitations

- Data is not stored permanently.
- No database integration.
- Only one mark value is stored per student.

12. Future Enhancements

- Add Roll Number support.
- Store multiple subject marks.
- Calculate percentage and grade.
- Save data to files or databases.
- Create a graphical user interface.

13. Conclusion

The Student Manager App is a simple and effective mini project for understanding Python fundamentals. It provides hands-on experience with dictionaries, loops, conditions, and menu-driven programming.